
FoMoD: Future Force-and-Motion Distillation Improves Contact-Rich Manipulation

Peiyuan Tang^{1,*} Yao Li^{2,*} Xiaodong Zhang^{2,†} Wuyang Zhang²
Qin Xia¹ Yanyong Zhang² Zijiang James Yang^{3,†}

¹School of Computer Science and Technology, Xi'an Jiaotong University

²School of Computer Science and Technology, University of Science and Technology of China

³Synkrotron, Inc.

*Equal contribution. †Corresponding authors.

Abstract

World-action models improve robotic manipulation by predicting future scene evolution to guide policy learning. However, existing approaches typically model such evolution through pixel-space video prediction, which entangles task-relevant dynamics with irrelevant lighting and texture. Moreover, visual prediction struggles to capture future contact information, thereby limiting its applicability to contact-rich tasks. Our key insight is that future force and motion provide a more physically grounded representation of interaction dynamics, yielding more informative cues for action prediction in contact-rich manipulation. However, directly predicting raw force and motion signals is both difficult and unnecessary, as it requires precise reconstruction of noisy, task-irrelevant details. To address this issue, we propose Force-and-Motion Distillation (FoMoD), a framework that distills future contact and motion dynamics into the policy. FoMoD adopts a teacher-student distillation framework, in which privileged teachers extract compact, action-relevant representations from future sensor signals and distill them into a student model. This design not only makes future dynamics easier to predict but also encourages the model to focus on control-relevant information rather than on irrelevant details. To further improve robustness and generalization, we introduce a curriculum-based strategy that progressively injects noise into future queries, thereby reducing the model’s reliance on future information. We evaluate FoMoD on five real-world contact-rich manipulation tasks. Experiments show that FoMoD achieves state-of-the-art performance, consistently improves task success rates, and generalizes effectively to novel environments.

1 Introduction

Humans are naturally capable of performing dexterous manipulation such as USB insertion and component assembly. However, these seemingly simple daily tasks remain highly challenging for robots. Contact-rich manipulation, particularly high-precision insertion, requires not only accurate motion planning and execution, but also the ability to perceive and respond to contact forces in real time, while reasoning over complex physical interaction dynamics. Although vision-based policies have achieved remarkable progress in general-purpose robotic manipulation [16, 4, 3], they still struggle with contact-rich tasks, as contact states and force magnitudes are often difficult to infer reliably from visual alone. This limitation has motivated recent studies to incorporate physical feedback, such as force and tactile sensing, to improve manipulation performance

Recent studies on force-aware contact-rich manipulation have primarily focused on designing more effective multimodal architectures to fully exploit force/tactile feedback. For example, ForceVLA [42]

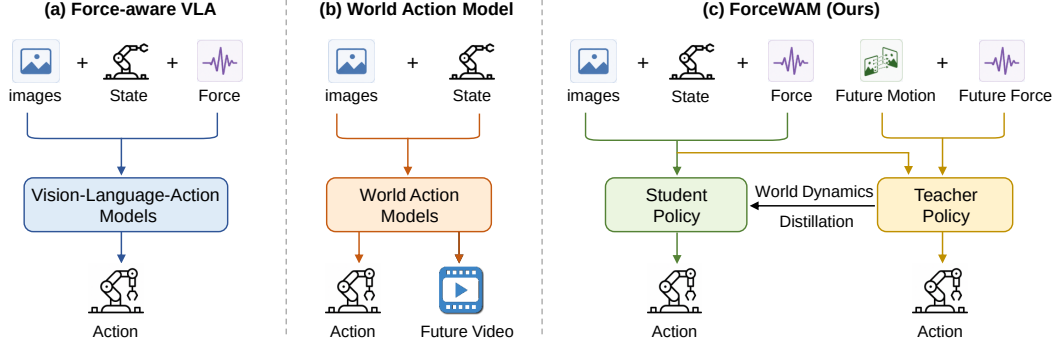


Figure 1: Comparison between baseline methods and FoMoD. Baselines directly map current observations to actions, which can cause modality collapse and limited generalization, while FoMoD distills future force and motion representations to make the student learn physical interaction dynamics that guide action generation.

employs a Mixture-of-Experts (MoE) architecture with modality-specific experts and uses dynamic routing to adaptively fuse visual and force features. TA-VLA [48] studies joint torque fusion in Vision-Language-Action(VLA) models and shows that injecting torque signals into the action generation module outperforms feeding them into the vision-language backbone. Other works [40, 50] investigate fast-slow dual-system architectures, where low-frequency vision provides global scene understanding, while high-frequency force/tactile feedback enables rapid contact response and closed-loop control. Nevertheless, these methods remain largely reactive, mapping current multimodal observations directly to actions without explicitly reasoning about future interaction dynamics.

A seemingly straightforward alternative is to directly extend existing video-based world action models (WAMs) to force-aware manipulation. However, this is difficult for three reasons. First, training such models requires large-scale, temporally aligned visual and force/tactile trajectories, whereas existing video corpora typically lack physical feedback and robot interaction data remain scarce. Second, contact-rich tasks demand high-frequency control, while explicit video prediction introduces substantial inference latency. Third, pixel-level video prediction can encourage models to focus on appearance details rather than the contact and motion dynamics most relevant to manipulation.

To address these challenges, we propose **FoMoD (Force-and-Motion Distillation)**, a physical-foresight framework for contact-rich manipulation, as illustrated in Figure 1. FoMoD leverages future force/torque and optical-flow representations as supervisory signals to learn interaction dynamics, which further guide action generation. By introducing future dynamics prediction as an auxiliary objective, FoMoD requires the model to extract force feedback and motion-relevant cues from current observations, rather than relying solely on visual semantics, thereby mitigating modality collapse and enabling explicit of physical dynamics.

To implement this idea, we employ a knowledge distillation [7, 39] framework. Instead of directly predicting high-dimensional raw future observations, which is computationally expensive and may force the model to capture redundant irrelevant details, we train a privileged teacher model that utilizes both current and future observations to extract compact latent dynamics features, which encode the most useful information about upcoming physical interactions. A student policy is then trained to match these latent representations using only current observations. By distilling such privileged future-aware dynamics into the student, the policy can implicitly capture future interaction patterns without requiring explicit future prediction or future signals at inference time. To further improve generalization, we introduce curriculum-based Progressive Noise Injection, which gradually perturbs the student’s latent queries during training.

We evaluate FoMoD on five real-world contact-rich manipulation tasks under both in-distribution and out-of-distribution settings. FoMoD achieves stronger task performance and better generalization to unseen visual and physical variations than existing VLA and force-aware baselines. Further ablations and representation analyses show that the student queries encode predictive information about future force and motion dynamics, indicating that FoMoD implicitly learns physical interaction dynamics rather than merely fitting expert actions. Our contributions are summarized as follows:

- We propose a future physical dynamics representation where force/torque reflects contact states and optical flow encodes motion dynamics, overcoming the limitations of purely visual representations in contact-rich manipulation.
- We introduce FoMoD, a force-and-motion distillation framework that transfers privileged future physical dynamics into a policy conditioned only on current observations, improving robustness and generalization to unseen objects and environmental variations.
- We empirically validate FoMoD on five real-world contact-rich manipulation tasks, demonstrating its effectiveness in improving task performance, out-of-distribution generalization, and implicit physical dynamics learning.

2 Related Work

Vision-Language-Action Models. Recent studies have leveraged pretrained vision-language models to build general-purpose robotic policies that follow open-ended instructions and perform diverse manipulation tasks [53, 4, 22, 3]. These methods encode visual observations and language instructions into multimodal representations and map them to robot actions. Existing approaches adapt VLMs into robot policies by either autoregressively predicting discretized action tokens [16, 28] or directly modeling continuous action distributions via flow matching [4, 3]. Building on these foundations, recent studies further improve VLA models by enhancing high-level reasoning and planning [21, 24], strengthening 3D spatial perception [30, 19], incorporating memory mechanisms [32, 38], and developing dual-system architectures [5, 33]. Despite these advances, most VLA policies rely primarily on visual observations and robot states, which provide limited information about contact events, force directions and friction. To bridge this gap, we introduce FoMoD, which fuses force/torque sensing with vision-language representations to improve precision and compliance in contact-rich manipulation.

Contact-Rich Manipulation with Physical Feedback. Several approaches have explored incorporating physical feedback into robotic policies. Forcemimic [23] and FoAR [10] introduce force-aware policies by integrating 6-DoF force/torque measurements, enabling robots to better perceive interaction dynamics. Other works [48, 27] exploit proprioceptive signals, such as joint efforts, to estimate external contact states without additional force sensors. Beyond force-based sensing, tactile-based approaches leverage high-resolution tactile images to capture localized contact geometry and surface interactions [13, 9, 14, 2]. To fuse these physical modalities with vision language features, existing methods adopt feature concatenation [45, 48, 9], cross-attention [11, 52, 17], gating mechanisms [10, 46, 50, 18], and Mixture-of-Experts architectures [15, 42, 20]. Building on this, recent works [43, 50, 25, 36] incorporate world models to reason about future physical interactions by jointly predict future force or tactile signals and future videos. However, this multimodal prediction paradigm requires large-scale synchronized visual and force/tactile data and incurs substantial computational overhead, leading to high inference latency. In contrast, our method distills task-relevant future force and motion signals into the policy, enable efficient inference by avoiding video generation.

World Models for VLA Policies. VLA policies learn a direct mapping from current visual-language observations to robot actions [53, 16, 4], whereas world models learn how future world states evolve through interaction [8, 51, 41]. Incorporating world modeling into VLA policies augments action supervision with future-state prediction, allowing action generation to account for future consequences rather than expert actions alone [12, 6, 47, 37]. Early world-action models realize this idea through future image or video prediction [51, 41], whereas latent world models replace expensive pixel-space rollouts with compact, task-relevant predictive representations. In contrast to approaches that derive action-oriented predictive representations primarily from future visual observations [49, 34, 26], FoMoD distills privileged future force/torque and optical-flow dynamics into the policy to capture contact physics that may be visually subtle or occluded, without requiring future inputs or explicit rollouts at inference time.

3 Method

We introduce **FoMoD**, a distillation framework that transfers future force and motion dynamics into vision-language-action models, enabling them to better model physical interactions. In this section,

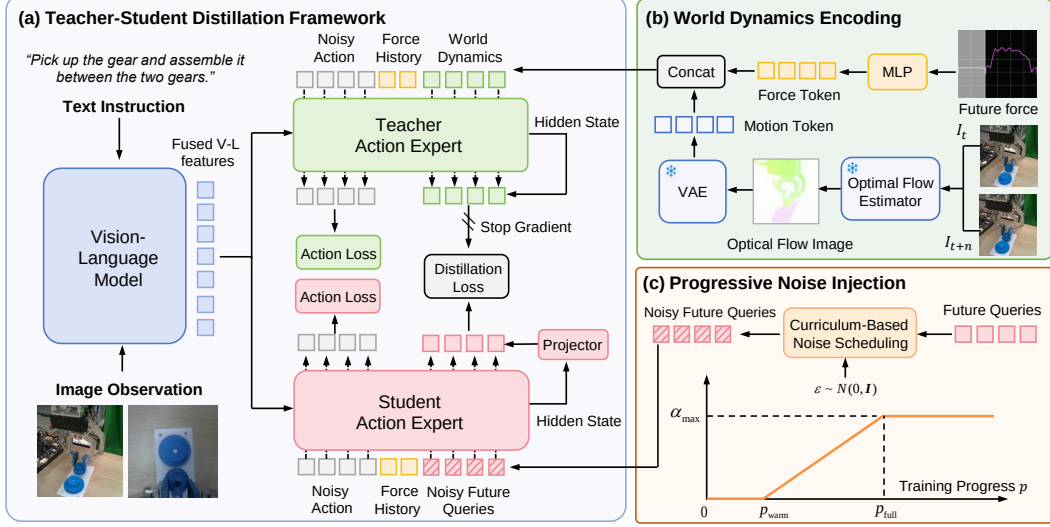


Figure 2: Overview of our proposed FoMoD. It consists of (a) a teacher-student distillation framework that transfers future-aware dynamics representations to the student policy, (b) a future dynamics encoder that converts future force and optical flow into force and motion tokens, and (c) progressive noise injection that adds scheduled noise to the student’s latent queries to improve generalization.

We begin with the problem formulation, followed by an overview of the framework, and then describe the training and inference strategies.

3.1 Problem Formulation and Overall Architecture

We formulate contact-rich manipulation as a policy learning problem. At each time step t , the policy π_θ takes the multimodal observation $o_t = (I_t, s_t, F_t, l)$ as input, and outputs an action chunk $\mathbf{a}_t = (a_t, \dots, a_{t+H-1})$ consisting of H actions to be executed sequentially:

$$\mathbf{a}_t = \pi_\theta(I_t, s_t, F_t, l). \quad (1)$$

where I_t denotes multi-view images, s_t denotes the robot state (end-effector pose or joint positions), F_t denotes 6D wrench measurements, and l denotes the language instruction. To train the policy, we collect a dataset consisting of N expert demonstrations, denoted as $\mathcal{D} = \{\tau_i\}_{i=1}^N$, where each trajectory is defined as $\tau_i = \{(o_t^i, a_t^i)\}_{t=1}^{T_i}$. The policy π_θ is optimized via behavior cloning [1, 31] by minimizing the discrepancy between predicted actions and expert demonstrated actions:

$$\mathcal{L}_{\text{BC}}(\theta) = \mathbb{E}_{(o_t, a_t) \sim \mathcal{D}} [\ell_{\text{act}}(a_t, \pi_\theta(o_t))], \quad (2)$$

where ℓ_{act} denotes the action-generation objective, which can be instantiated as either an ℓ_1 loss or a mean squared error (MSE) loss.

Overall Architecture. As illustrated in Figure 2, FoMoD augments a π_0 -style VLA policy with an online privileged teacher. A shared VLM backbone encodes the current images, robot state, force history, and language instruction into a multimodal context. Two action experts with identical architectures operate on this context but receive different information. The *student* is conditioned only on current observations, whereas the *teacher* additionally receives compact tokens encoding future force and motion dynamics. These privileged cues allow the teacher to form future-aware latent representations of the upcoming physical interaction. Through teacher–student distillation, the student is trained to infer corresponding physical-dynamics representations from current observations alone. Both branches are jointly optimized for action generation, while latent alignment transfers the teacher’s future-aware knowledge to the student. At inference time, the future-dynamics encoder and teacher branch are discarded, and only the student policy is deployed.

3.2 Privileged Future Dynamics Encoding

Choosing the right future modality is critical for contact-rich manipulation. Prior approaches [12, 37, 44] often predict future visual observations, but RGB frames mainly describe appearance and are insufficient to capture physical interaction, such as contact onset, resistance, sliding, or jamming. We therefore use future force/torque measurements and optical flow as privileged dynamics cues. Force directly reflects contact dynamics, while optical flow captures how the robot and objects move through the scene. Together, they provide compact supervision that is more aligned with physical control than reconstructing future RGB frames.

Future Force Encoding. Given a future horizon of K steps, we extract the 6D wrench sequence $\mathbf{F}t + K$ from the force/torque sensor. This sequence characterizes the evolution of physical interaction over time, including contact onset, force direction and magnitude, and the response induced by environmental constraints. We encode the sequence with an MLP to obtain a compact force token:

$$\mathbf{f}_t = \text{MLP}(\mathbf{F}_{t:t+K}). \quad (3)$$

Future Motion Encoding. Optical flow provides a dense representation of scene motion by describing the pixel-wise displacement between two frames. Given the current image I_t and a future image I_{t+H} , we use a pretrained optical-flow estimation model [35] to obtain a dense motion field. For each pixel p , the corresponding flow vector $\mathbf{u}(p) = (u_x(p), u_y(p))$ represents its horizontal and vertical displacement in image space, thereby capturing the image-space motion of the robot and manipulated objects.

To align the optical flow with the image representation used in our framework, we colorize the two-channel flow field into a three-channel motion image $\mathbf{M}_t$. Specifically, we map the flow direction to hue, set saturation to one, and encode the normalized and clipped flow magnitude as value before converting the resulting HSV representation into RGB. We then compress $\mathbf{M}_t$ using a pretrained image VAE and employ a fixed set of learnable motion queries Q_m to aggregate the encoded features into compact motion tokens:

$$\mathbf{m}_t = \text{CrossAttn}(Q_m, \text{VAE}(\mathbf{M}_t)). \quad (4)$$

3.3 Online Future-Dynamics Distillation

Directly predicting future sensory observations imposes a difficult learning objective and may allocate model capacity to details that are irrelevant to control. FoMoD instead distills compact latent dynamics, allowing a privileged teacher to identify future information that is useful for action generation and transfer it to a causal student. The student can therefore learn predictive physical representations from current observations without reconstructing future signals or requiring them at deployment. We optimize the teacher and student jointly so that the distilled targets remain aligned with the evolving action objective.

The student uses a set of learnable queries $\mathbf{Q} \in \mathbb{R}^{N_q \times D}$ to predict the teacher’s future-aware intermediate representations from current observations. Let $\mathcal{L}$ denote the set of intermediate layers selected for alignment, and let $y, h \in \mathbb{R}^{|\mathcal{L}| \times N_q \times D}$ denote the teacher and student query representations extracted from these layers, respectively. Because the two representations have the same dimensions, we directly minimize their cosine distance:

$$\mathcal{L}_{\text{distill}} = \frac{1}{|\mathcal{L}| \cdot N_q} \sum_{l \in \mathcal{L}} \sum_{i=1}^{N_q} [1 - \cos(h^{(l,i)}, \text{sg}(y^{(l,i)}))], \quad (5)$$

where $\text{sg}(\cdot)$ stops gradients through the teacher target. Without stop-gradient, both branches could jointly reduce the alignment loss by drifting toward uninformative representations, weakening the privileged supervision and potentially causing representation collapse.

3.4 Joint Training and Inference

Progressive Query Perturbation. Future interaction dynamics are inherently stochastic, and directly matching deterministic teacher representations may cause the student to overfit privileged future information. Let $\mathbf{Q}$ denote the original student queries, $\tilde{\mathbf{Q}}$ their perturbed counterparts,

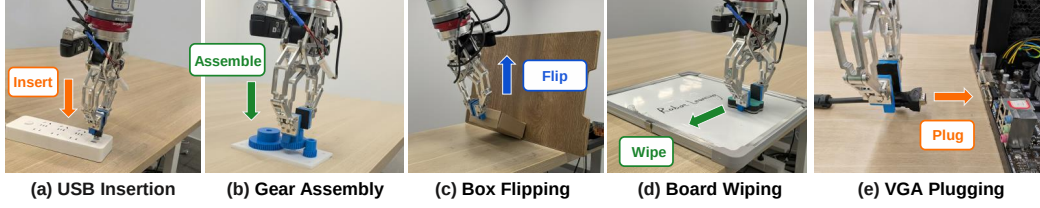


Figure 3: Five real-world contact-rich manipulation tasks used in our experiments.

$p \in [0, 1]$ the normalized training progress, $\sigma(p)$ the corresponding noise scale, and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ Gaussian noise with the same shape as $\mathbf{Q}$. We model future uncertainty and reduce this dependence by perturbing the queries as:

$$\tilde{\mathbf{Q}} = \mathbf{Q} + \sigma(p)\epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (6)$$

The noise scale $\sigma(p)$ is zero during warmup, increases linearly, and remains constant after reaching its maximum. This curriculum first allows the student to learn a meaningful alignment target and then encourages it to infer future-aware features robustly from causal observations.

Action Learning via Flow Matching. We train action prediction for both the student and teacher policies using flow matching. Given an expert action chunk $\mathbf{a}$, we sample $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\tau \sim \mathcal{U}(0, 1)$, construct $\mathbf{a}_\tau = (1 - \tau)\epsilon + \tau\mathbf{a}$, and define the target velocity as $v^* = \mathbf{a} - \epsilon$. Conditioned on c_S and c_T , the student and teacher losses are

$$\mathcal{L}_S = \mathbb{E}[\|v_S(\mathbf{a}_\tau, \tau, c_S) - v^*\|_2^2], \quad \mathcal{L}_T = \mathbb{E}[\|v_T(\mathbf{a}_\tau, \tau, c_T) - v^*\|_2^2]. \quad (7)$$

The student and teacher share the same noise sample and interpolation time, ensuring that their velocity predictions correspond to the same interpolation state. The final training objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_S + \mathcal{L}_T + \lambda \mathcal{L}_{\text{distill}}, \quad (8)$$

where $\lambda = 0.5$ balances action learning and representation alignment.

Inference. At inference, the teacher branch, including the future-dynamics encoder and teacher action expert, is discarded, and actions are generated solely by the student policy. The deployed model therefore follows the same causal observation-to-action pipeline as standard VLA policies, requiring no future force or motion inputs and introducing negligible additional inference latency.

4 Experiments

In this section, we conduct extensive experiments to evaluate the effectiveness of our method on contact-rich manipulation tasks. Our experiments aim to answer the following key questions:

- Q1:** Does FoMoD improve both in-distribution and out-of-distribution performance?
- Q2:** Which future-dynamics representation best supports contact-rich manipulation: future visual observations, force signals, motion cues, or their combination?
- Q3:** Does the student model learn to infer future physical interactions in latent space?

4.1 Experimental Setup

Hardware Platform and Data Collection We conduct real-world experiments on a 7-DoF xArm7 robotic manipulator equipped with a wrist-mounted Orbbec Gemini 336L camera, a fixed third-person RealSense L515 camera, and a 6D force/torque sensor. We teleope the robot using a 3D SpaceMouse and collect data at 30Hz. We record only successful trajectories, including synchronized visual observations, robot states, force-torque readings, and actions, and use them for model training.

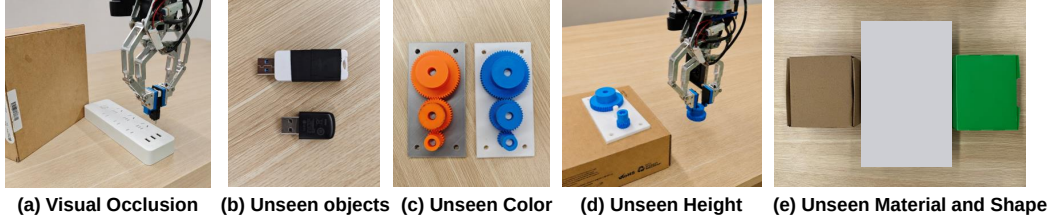


Figure 4: Out-of-distribution generalization settings used in our real-world evaluation.

Table 1: Performance comparison on In-Distribution (ID) contact-rich manipulation tasks. We report the success rate (%) across different tasks. Avg. denotes the average success rate across all ID evaluation settings. Best results are highlighted in **bold**.

Model	USB Insertion	Gear Assembly	Box Flipping	Board Wiping	VGA Plugging	Avg.
π_0 [4] w/o force	44.0	46.7	55.0	60.0	0.0	41.1
π_0 [4] w/ force	64.0	83.3	70.0	70.0	5.0	58.5
TA-VLA [48]	80.0	93.3	80.0	75.0	10.0	67.7
ForceVLA [42]	52.0	66.7	65.0	65.0	5.0	50.7
Seer [37] w/ Force	76.0	70.0	95.0	70.0	50.0	72.2
Ours	92.0	83.3	95.0	80.0	85.0	87.1

Tasks and Evaluation Metrics. We evaluate our method on five real-world challenging contact-rich manipulation tasks: USB Insertion, Gear Assembly, Box Flipping, Board Wiping, and VGA Plugging, as shown in Figure 3. These tasks requires diverse contact-rich interaction patterns, including precise insertion, rigid-body assembly, object reorientation, and sustained surface contact. The detailed For each task, we collect 80–100 successful demonstration trajectories and evaluate each method with more than 20 real-world rollouts. We report task success rate as the primary metric. Detailed task descriptions and success criteria are provided in Appendix D.

Compared Baselines. We compare our method against several strong baselines. π_0 [4] **w/o Force** is a flow-matching-based Vision-Language-Action model pretrained on large-scale robot data. π_0 **w/ Force** is a force-augmented variant of π_0 that converts force measurements into tokens and concatenates them with the input sequence. **TA-VLA** [48] is a force-aware VLA model that introduces an auxiliary future-force prediction objective. **ForceVLA** [42] is a force-aware VLA model that adopts a Mixture-of-Experts architecture [15] to model visual and force modalities with specialized experts. **Seer** [37] **w/ Force** is a force-aware variant of Seer that uses force input but directly predicts future visual observations as its auxiliary supervision. For a fair comparison, all methods are fully fine-tuned on our collected dataset.

Implementation Details Our model consists of a pretrained 2.7B PaliGemma VLM and two action experts, including a 300M teacher expert and a 300M student expert. The model takes observations from a fixed third-person camera and a wrist-mounted camera, along with 6D force/torque measurements over 10 historical steps. We generate future optical flow with a pretrained RAFT [35] model and encode it using a pretrained SDXL-VAE [29]. The number of future queries is set to 16 for optical flow and 1 for future force. For distillation, we align the intermediate representations of the teacher and student action experts at layers $\mathcal{A} = \{8, 12, 16\}$. For action prediction, Our model outputs a 32-step action chunk of delta end-effector poses. During training, we fully fine-tune a multi-task model for 30K steps on a single A100-80G GPU with a batch size of 16. Training is performed using the AdamW optimizer with a cosine learning-rate schedule, including 1,000 warmup steps, with the learning rate decaying from 2.5×10^{-5} to 2.5×10^{-6} .

4.2 Main Results.

Table 1 summarizes the performance on in-distribution contact-rich manipulation tasks. FoMoD achieves the best overall performance, demonstrating that modeling future physical dynamics provides

Table 2: Generalization performance on Out-of-Distribution (OOD) contact-rich manipulation tasks. We report the success rate (%). Avg. denotes the average success rate across all OOD evaluation settings. Best results are highlighted in **bold**.

Model	USB Insertion		Gear Assembly		Box Flipping		Avg.
	Object	Occlusion	Height	Color	Material	Shape	
π_0 [4] w/o force	20.0	26.7	0.0	40.0	26.7	0.0	18.9
π_0 [4] w/ force	40.0	73.3	0.0	46.7	26.7	33.3	36.7
TA-VLA [48]	20.0	66.7	13.3	66.7	13.3	26.7	34.5
ForceVLA [42]	26.7	20.0	0.0	40.0	0.0	16.7	17.2
Seer [37] w/ Force	46.7	20.0	53.3	60.0	73.3	40.0	48.9
Ours	86.7	86.7	46.7	60.0	80.0	53.3	68.9

Table 3: Ablation study of visual, force, and motion representations of future dynamics. We report success rates (%) on USB Insertion, Box Flipping, and VGA Plugging.

Future Target			Success Rate (%)			
Visual	Force	Motion	USB Insertion	Box Flipping	VGA Plugging	Avg.
–	–	✓	92.0	60.0	60.0	70.7
–	✓	–	84.0	90.0	30.0	68.0
✓	✓	–	80.0	85.0	35.0	66.7
–	✓	✓	92.0	95.0	85.0	90.7

effective guidance for contact-rich manipulation. The improvement is most significant on VGA Plugging, where FoMoD reaches a success rate of 85.0%, compared with 50.0% achieved by the strongest baseline. This result suggests that FoMoD enables more fine-grained correction during precise insertion. FoMoD also clearly outperforms Seer, indicating that predicting future visual observations alone is insufficient for high-precision contact-rich manipulation. The main exception is Gear Assembly, where FoMoD performs worse than TA-VLA. This is mainly because FoMoD shows limited spatial generalization in this task and sometimes fails to pick the gear at boundary positions.

4.3 Out-of-Distribution Generalization

OOD Evaluation Settings. We evaluate OOD generalization on three representative contact-rich tasks, as shown in Figure 4. For USB Insertion, we consider *visual occlusion*, where the external camera view is partially blocked by an object, and *unseen objects*, where the USB connector is replaced with a novel instance. For Gear Assembly, we consider *unseen color*, which changes the object appearance, and *unseen height*, which changes the target assembly height. For Box Flipping, we consider *unseen material*, where the original wooden box is replaced with a 3D-printed box, leading to different friction properties, and *unseen shape*, which evaluates the policy’s ability to flip boxes with different geometries. All OOD settings keep the original task objective and success criterion unchanged, thereby measuring robustness to visual and physical distribution shifts rather than adaptation to new task definitions. Each OOD setting is evaluated over 15 trials.

Results. Table 2 reports the OOD evaluation results. FoMoD achieves the best average success rate of 68.9%, outperforming the strongest baseline, Seer w/ Force, by 20.0 percentage points. The gains are most pronounced on USB Insertion, where FoMoD reaches 86.7% under both object variation and occlusion, and on Box Flipping, where it performs best under both material and shape shifts. These results suggest that distilling future force-and-motion cues helps the policy handle changes in object appearance, visibility, and contact response. Gear Assembly shows a more mixed trend: FoMoD improves over most baselines but remains slightly worse than Seer w/ Force under height changes and TA-VLA under color changes, indicating that fine spatial alignment under visual or geometric shifts remains challenging. Overall, FoMoD improves OOD generalization by learning future-aware latent dynamics that remain useful beyond the training distribution.

Table 4: Architecture ablation of the teacher–student distillation design. USB Insertion and Box Flipping are evaluated under both in-distribution (ID) and out-of-distribution (OOD) settings. Avg. is computed across all reported settings.

Configuration	USB Insertion		Box Flipping		VGA Plugging	Avg.
	ID	OOD	ID	OOD		
w/o Distill. Loss	44.0	30.0	85.0	30.0	25.0	42.8
w/o Teacher Loss	60.0	36.7	80.0	16.7	10.0	40.7
w/o Stop Gradient	68.0	50.0	60.0	16.7	20.0	42.9
Full Model	92.0	86.7	95.0	66.7	85.0	85.1

Table 5: Ablation of the training strategies used in FoMoD. USB Insertion and Box Flipping are evaluated under both in-distribution (ID) and out-of-distribution (OOD) settings. Avg. is computed across all reported settings.

Training Strategy	USB Insertion		Box Flipping		VGA Plugging	Avg.
	ID	OOD	ID	OOD		
Joint Prediction	84.0	76.7	65.0	20.0	65.0	62.1
w/o Progressive Noise	84.0	73.3	90.0	40.0	85.0	74.1
Full Model	92.0	86.7	95.0	66.7	85.0	85.1

4.4 Ablation Studies

Ablation on Future Prediction Targets. Table 3 studies the role of visual, force, and motion representations of future dynamics, instantiated as RGB observations, force signals, and optical flow, respectively. Future force is particularly useful for tasks that require sustained contact regulation, such as Box Flipping, where force-only supervision already achieves strong performance. Future optical flow provides more direct cues for motion and alignment, which benefits tasks that require repeated local adjustment and probing, such as USB Insertion and VGA Plugging. Combining force and optical flow gives the best overall performance, showing that the two modalities capture complementary aspects of future interaction dynamics. In contrast, adding future RGB provides little benefit and even degrades performance, suggesting that raw visual prediction introduces redundant appearance details that are less action-relevant than optical flow. This supports our design choice of using future force and optical flow as the privileged future representation.

Ablation on Model Architecture. Table 4 isolates the architectural components of the teacher–student distillation design. Removing the distillation loss causes the average success rate to drop from 85.1% to 42.8%, showing that future-aware teacher representations are not transferred to the student through action supervision alone. Removing the teacher action loss yields a similarly large degradation, indicating that the privileged branch must remain grounded in action generation to provide useful distillation targets. Performance also drops substantially without stop-gradient, supporting the use of a fixed directional target that prevents the teacher representation from drifting toward the student. Together, these results validate the asymmetric teacher–student architecture used by FoMoD.

Ablation on Training Strategies. Table 5 evaluates joint prediction and progressive noise injection separately from the distillation architecture. The joint-prediction variant reaches an average success rate of 62.1%, substantially below the full model, indicating that directly adding predictive supervision is less effective than the complete FoMoD training scheme. Removing progressive noise reduces the average success rate from 85.1% to 74.1%. The degradation is larger under OOD shifts than in the corresponding ID settings, suggesting that progressive noise regularizes the student representation and improves its transfer to unseen visual and physical conditions.

5 Conclusion

We presented **FoMoD**, a force-and-motion distillation framework for contact-rich manipulation. Rather than predicting high-dimensional future observations or simply appending force readings to the policy input, FoMoD learns a compact future-dynamics representation from privileged future force/torque signals and optical flow. Through teacher–student distillation, this future-aware representation is transferred to a deployable student policy that only uses current observations during inference. We further introduced progressive noise injection to regularize the distillation process and improve the student’s ability to infer future dynamics from causal inputs. Experiments on five real-world contact-rich manipulation tasks show that FoMoD improves both in-distribution performance and out-of-distribution generalization over general-purpose and force-aware VLA baselines. Ablation studies further verify the complementary roles of future force and optical flow, as well as the benefit of progressive noise injection. Overall, these results demonstrate that distilling future-aware physical dynamics into latent policy representations is an effective approach to improving action generation in contact-rich robotic manipulation.

References

- [1] Michael Bain and Claude Sammut. A framework for behavioural cloning. In *Machine intelligence 15*, pages 103–129, 1995.
- [2] Jianxin Bi, Kevin Yuchen Ma, Ce Hao, Mike Zheng Shou, and Harold Soh. Vla-touch: Enhancing vision-language-action models with dual-level tactile feedback. *arXiv preprint arXiv:2507.17294*, 2025.
- [3] Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi Fan, Yu Fang, Dieter Fox, Fengyuan Hu, Spencer Huang, et al. Gr00t n1: An open foundation model for generalist humanoid robots. *arXiv preprint arXiv:2503.14734*, 2025.
- [4] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. π_0 : A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.
- [5] Hao Chen, Jiaming Liu, Chenyang Gu, Zhuoyang Liu, Renrui Zhang, Xiaoqi Li, Xiao He, Yandong Guo, Chi-Wing Fu, Shanghang Zhang, et al. Fast-in-slow: A dual-system foundation model unifying fast manipulation within slow reasoning. *arXiv preprint arXiv:2506.01953*, 2025.
- [6] Chelsea Finn and Sergey Levine. Deep visual foresight for planning robot motion. In *2017 IEEE international conference on robotics and automation (ICRA)*, pages 2786–2793. IEEE, 2017.
- [7] Jianping Gou, Baosheng Yu, Stephen J Maybank, and Dacheng Tao. Knowledge distillation: A survey. *International journal of computer vision*, 129(6):1789–1819, 2021.
- [8] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. *arXiv preprint arXiv:1912.01603*, 2019.
- [9] Peng Hao, Chaofan Zhang, Dingzhe Li, Xiaoge Cao, Xiaoshuai Hao, Shaowei Cui, and Shuo Wang. Tla: Tactile-language-action model for contact-rich manipulation. *arXiv preprint arXiv:2503.08548*, 2025.
- [10] Zihao He, Hongjie Fang, Jingjing Chen, Hao-Shu Fang, and Cewu Lu. Foar: Force-aware reactive policy for contact-rich robotic manipulation. *IEEE Robotics and Automation Letters*, 2025.
- [11] Liang Heng, Haoran Geng, Kaifeng Zhang, Pieter Abbeel, and Jitendra Malik. Vitacformer: Learning cross-modal representation for visuo-tactile dexterous manipulation. *arXiv preprint arXiv:2506.15953*, 2025.
- [12] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. *arXiv preprint arXiv:2412.14803*, 2024.

- [13] Binghao Huang, Yixuan Wang, Xinyi Yang, Yiyue Luo, and Yunzhu Li. 3d-vitac: Learning fine-grained manipulation with visuo-tactile sensing. *arXiv preprint arXiv:2410.24091*, 2024.
- [14] Jialei Huang, Shuo Wang, Fanqi Lin, Yihang Hu, Chuan Wen, and Yang Gao. Tactile-vla: unlocking vision-language-action model’s physical knowledge for tactile generalization. *arXiv preprint arXiv:2507.09160*, 2025.
- [15] Robert A Jacobs, Michael I Jordan, Steven J Nowlan, and Geoffrey E Hinton. Adaptive mixtures of local experts. *Neural computation*, 3(1):79–87, 1991.
- [16] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.
- [17] Geonhyup Lee, Yeongjin Lee, Kangmin Kim, Seongju Lee, Sangjun Noh, Seunghyeok Back, and Kyoobin Lee. Manipforce: Force-guided policy learning with frequency-aware representation for contact-rich manipulation. *arXiv preprint arXiv:2509.19047*, 2025.
- [18] Jiuzhou Lei, Chang Liu, Yu She, Xiao Liang, and Minghui Zheng. Learning when to see and when to feel: Adaptive vision-torque fusion for contact-aware manipulation. *arXiv preprint arXiv:2604.01414*, 2026.
- [19] Fuhao Li, Wenxuan Song, Han Zhao, Jingbo Wang, Pengxiang Ding, Donglin Wang, Long Zeng, and Haoang Li. Spatial forcing: Implicit spatial representation alignment for vision-language-action model. *arXiv preprint arXiv:2510.12276*, 2025.
- [20] Yang Li, Hongru Jiang, Junjie Xia, Hongquan Zhang, Jinda Du, Yunsong Zhou, Jia Zeng, Ce Hao, Jieji Ren, Qiaojun Yu, et al. Forcevla2: Unleashing hybrid force-position control with force awareness for contact-rich manipulation. *arXiv preprint arXiv:2603.15169*, 2026.
- [21] Fanqi Lin, Ruiqian Nai, Yingdong Hu, Jiacheng You, Junming Zhao, and Yang Gao. Onetwovla: A unified vision-language-action model with adaptive reasoning. *arXiv preprint arXiv:2505.11917*, 2025.
- [22] Songming Liu, Lingxuan Wu, Bangguo Li, Hengkai Tan, Huayu Chen, Zhengyi Wang, Ke Xu, Hang Su, and Jun Zhu. Rdt-1b: a diffusion foundation model for bimanual manipulation. *arXiv preprint arXiv:2410.07864*, 2024.
- [23] Wenhai Liu, Junbo Wang, Yiming Wang, Weiming Wang, and Cewu Lu. Forcemimic: Force-centric imitation learning with force-motion capture system for contact-rich manipulation. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 1105–1112. IEEE, 2025.
- [24] Zhuoyang Liu, Jiaming Liu, Hao Chen, Jiale Yu, Ziyu Guo, Chengkai Hou, Chenyang Gu, Xiangju Mi, Renrui Zhang, Kun Wu, et al. Last_{0}: Latent spatio-temporal chain-of-thought for robotic vision-language-action model. *arXiv preprint arXiv:2601.05248*, 2026.
- [25] Yunfan Lou, Yifan Ye, Yankai Fu, Jun Cen, Xiaowei Chi, Yaoxu Lyu, Peidong Jia, Sirui Han, Zhihe Lu, and Shanghang Zhang. Dream-tac: A unified tactile world action model for contact-rich robot manipulation. 2026.
- [26] Hao Luo, Wanpeng Zhang, Yicheng Feng, Sipeng Zheng, Haiweng Xu, Chaoyi Xu, Ziheng Xi, Yuhui Fu, and Zongqing Lu. Being-h0.7: A latent world-action model from egocentric videos. 2026.
- [27] Steven Oh, Jason Jingzhou Liu, Tony Tao, Philip Han, Kenneth Shaw, Satoshi Funabashi, Ruslan Salakhutdinov, and Deepak Pathak. Factr 2: Learning external force sensing for commodity robot arms improves policy learning. 2026.
- [28] Karl Pertsch, Kyle Stachowicz, Brian Ichter, Danny Driess, Suraj Nair, Quan Vuong, Oier Mees, Chelsea Finn, and Sergey Levine. Fast: Efficient action tokenization for vision-language-action models. *arXiv preprint arXiv:2501.09747*, 2025.

- [29] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. Sdxl: Improving latent diffusion models for high-resolution image synthesis. *arXiv preprint arXiv:2307.01952*, 2023.
- [30] Delin Qu, Haoming Song, Qizhi Chen, Yuanqi Yao, Xinyi Ye, Yan Ding, Zhigang Wang, JiaYuan Gu, Bin Zhao, Dong Wang, et al. Spatialvla: Exploring spatial representations for visual-language-action model. *arXiv preprint arXiv:2501.15830*, 2025.
- [31] Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics*, pages 627–635. JMLR Workshop and Conference Proceedings, 2011.
- [32] Hao Shi, Bin Xie, Yingfei Liu, Lin Sun, Fengrong Liu, Tiancai Wang, Erjin Zhou, Haoqiang Fan, Xiangyu Zhang, and Gao Huang. Memoryvla: Perceptual-cognitive memory in vision-language-action models for robotic manipulation. *arXiv preprint arXiv:2508.19236*, 2025.
- [33] Haoming Song, Delin Qu, Yuanqi Yao, Qizhi Chen, Qi Lv, Yiwen Tang, Modi Shi, Guanghui Ren, Maoqing Yao, Bin Zhao, et al. Hume: Introducing system-2 thinking in visual-language-action model. *arXiv preprint arXiv:2505.21432*, 2025.
- [34] Yue Su, Sijin Chen, Haixin Shi, Mingyu Liu, Zhengshen Zhang, Ningyuan Huang, Weiheng Zhong, Zhengbang Zhu, Yuxiao Liu, and Xihui Liu. World guidance: World modeling in condition space for action generation. *arXiv preprint arXiv:2602.22010*, 2026.
- [35] Zachary Teed and Jia Deng. Raft: Recurrent all-pairs field transforms for optical flow. In *European conference on computer vision*, pages 402–419. Springer, 2020.
- [36] Shuai Tian, Yupeng Zheng, Yuhang Zheng, Songen Gu, Yujie Zang, Yuxin Qin, Weize Li, Haoran Li, Wenchao Ding, and Dongbin Zhao. Vt-wam: Visual-tactile world action model for contact-rich manipulation. 2026.
- [37] Yang Tian, Sizhe Yang, Jia Zeng, Ping Wang, Dahua Lin, Hao Dong, and Jiangmiao Pang. Predictive inverse dynamics models are scalable learners for robotic manipulation. *arXiv preprint arXiv:2412.15109*, 2024.
- [38] Marcel Torne, Karl Pertsch, Homer Walke, Kyle Vedder, Suraj Nair, Brian Ichter, Allen Z Ren, Haohuan Wang, Jiaming Tang, Kyle Stachowicz, et al. Mem: Multi-scale embodied memory for vision language action models. *arXiv preprint arXiv:2603.03596*, 2026.
- [39] Lin Wang and Kuk-Jin Yoon. Knowledge distillation and student-teacher learning for visual intelligence: A review and new outlooks. *IEEE transactions on pattern analysis and machine intelligence*, 44(6):3048–3068, 2021.
- [40] Han Xue, Jieji Ren, Wendi Chen, Gu Zhang, Yuan Fang, Guoying Gu, Huazhe Xu, and Cewu Lu. Reactive diffusion policy: Slow-fast visual-tactile policy learning for contact-rich manipulation. *arXiv preprint arXiv:2503.02881*, 2025.
- [41] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, et al. World action models are zero-shot policies. *arXiv preprint arXiv:2602.15922*, 2026.
- [42] Jiawen Yu, Hairuo Liu, Qiaojun Yu, Jieji Ren, Ce Hao, Haitong Ding, Guangyu Huang, Guofan Huang, Yan Song, Panpan Cai, et al. Forcevla: Enhancing vla models with a force-aware moe for contact-rich manipulation. *arXiv preprint arXiv:2505.22159*, 2025.
- [43] Yujie Zang, Yuhang Zheng, Xiangli Nie, Yupeng Zheng, Shuai Tian, Songen Gu, Chen Gao, Zining Wang, Shuicheng Yan, and Wenchao Ding. Tacforesight: Force-guided tactile world model for contact-rich manipulation. 2026.
- [44] Jia Zeng, Qingwen Bu, Bangjun Wang, Wenke Xia, Li Chen, Hao Dong, Haoming Song, Dong Wang, Di Hu, Ping Luo, et al. Learning manipulation by predicting interaction. *arXiv preprint arXiv:2406.00439*, 2024.

- [45] Chaofan Zhang, Peng Hao, Xiaoge Cao, Xiaoshuai Hao, Shaowei Cui, and Shuo Wang. Vtla: Vision-tactile-language-action model with preference learning for insertion manipulation. *arXiv preprint arXiv:2505.09577*, 2025.
- [46] Kaidi Zhang, Heng Zhang, Zhengtong Xu, Zhiyuan Zhang, Md Rakibul Islam Prince, Xiang Li, Xiaojing Han, Yuhao Zhou, Arash Ajoudani, and Yu She. Tacvla: Contact-aware tactile fusion for robust vision-language-action manipulation. *arXiv preprint arXiv:2603.12665*, 2026.
- [47] Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, Xinqiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, Fan Lu, He Wang, et al. Dreamvla: a vision-language-action model dreamed with comprehensive world knowledge. *arXiv preprint arXiv:2507.04447*, 2025.
- [48] Zongzheng Zhang, Haobo Xu, Zhuo Yang, Chenghao Yue, Zehao Lin, Huan-ang Gao, Ziwei Wang, and Hao Zhao. Ta-vla: Elucidating the design space of torque-aware vision-language-action models. *arXiv preprint arXiv:2509.07962*, 2025.
- [49] Ruijie Zheng, Jing Wang, Scott Reed, Johan Bjorck, Yu Fang, Fengyuan Hu, Joel Jang, Kaushil Kundalia, Zongyu Lin, Loic Magne, et al. Flare: Robot learning with implicit world modeling. *arXiv preprint arXiv:2505.15659*, 2025.
- [50] Yuhang Zheng, Songen Gu, Weize Li, Yupeng Zheng, Yujie Zang, Shuai Tian, Xiang Li, Ce Hao, Chen Gao, Si Liu, et al. Omnivta: Visuo-tactile world modeling for contact-rich robotic manipulation. *arXiv preprint arXiv:2603.19201*, 2026.
- [51] Chuning Zhu, Raymond Yu, Siyuan Feng, Benjamin Burchfiel, Paarth Shah, and Abhishek Gupta. Unified world models: Coupling video and action diffusion for pretraining on large robotic datasets. *arXiv preprint arXiv:2504.02792*, 2025.
- [52] Xinyue Zhu, Binghao Huang, and Yunzhu Li. Touch in the wild: Learning fine-grained manipulation with a portable visuo-tactile gripper. *arXiv preprint arXiv:2507.15062*, 2025.
- [53] Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning*, pages 2165–2183. PMLR, 2023.

A Limitations and Future Work

Our method has two main limitations. First, FoMoD relies on an external optical-flow estimator to construct future motion cues during training. Therefore, the learned latent dynamics may be affected by the quality of the estimated flow, especially under motion blur, occlusion. More accurate or task-adaptive motion estimators may further improve the quality of the distilled dynamics representation. Second, our experiments are conducted on a single VLA architecture, namely a π_0 -style policy with a flow-matching action head. Although our framework is architecture-agnostic, its effectiveness on other policy architectures, such as diffusion-based policies, autoregressive action-token models, or other VLA backbones, remains to be validated. Extending FoMoD to broader robot policy architectures is an important direction for future work.

Broader Impact. Force-aware contact-rich manipulation may benefit robotic assembly, maintenance, and assistive manipulation by improving robustness in tasks where physical feedback is important. At the same time, learned contact policies should be deployed with conservative force limits and safety monitors, since incorrect contact reasoning can damage objects, fixtures, or robot hardware.

B Data Collection

Figure 5 shows the real-world experimental setup and representative task initializations used for data collection and evaluation. Before each rollout, we reset the robot, manipulated object, and task fixture to the corresponding initial configuration, and verify that the target region is visible from the camera views. The demonstrations are collected by teleoperation with the same workspace layout and sensing setup used during policy evaluation. For each task, the initial pose preserves the intended contact-rich challenge while avoiding unreachable or unsafe starting states. During both demonstration collection and policy evaluation, the robot starts from the same task-specific reset region, and the manipulated object and target fixture are placed according to the corresponding initialization shown in the figure.

C Details of Compared Baselines

Seer+Force. This baseline tests whether the Seer architecture can benefit from force sensing through direct multimodal fusion alone. We augment Seer with encoded 6-axis wrist force/torque tokens and concatenate them with the original multimodal inputs before action prediction. The training objective is kept unchanged, so this comparison isolates the effect of adding force observations without introducing structured future-dynamics supervision.

π_0 w/o force. We adopt π_0 as the primary generic VLA baseline because it is a strong pretrained vision-language-action policy with a flow-matching action head. For our experiments, π_0 takes the same visual observations, language instruction, and robot proprioceptive state as our method, but it does not receive force measurements and does not include any explicit future-dynamics modeling module. The model is fine-tuned end-to-end on our task-specific demonstrations and predicts the same action-chunk format as all other methods.

π_0 w/ force. This baseline tests whether simply exposing a strong VLA model to force signals is sufficient for contact-rich manipulation. We extend π_0 by encoding the recent 6-axis force/torque history into force tokens and concatenating them with the multimodal input sequence. The action head

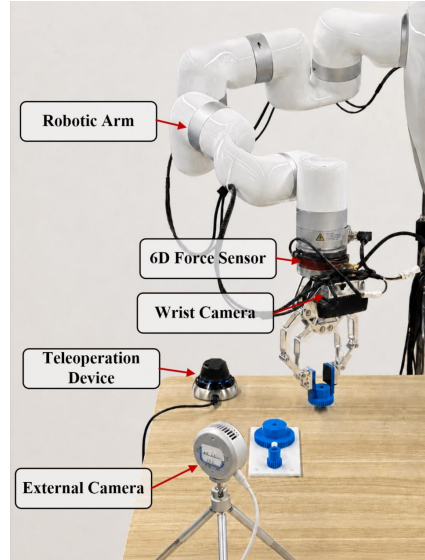


Figure 5: Real-world experimental setup and task initialization examples.

and training objective remain unchanged. This comparison isolates the benefit of force observation alone from the benefit of structured predictive dynamics learning.

TA-VLA. TA-VLA is included as a representative force-aware VLA baseline that uses an auxiliary future-prediction objective during training. In our setting, the model consumes the same causal inputs as our method and is fine-tuned on the same dataset split. Unlike our method, TA-VLA does not decompose policy learning into separate semantic planning, force prediction, and action generation stages. Instead, future-force supervision is used as an auxiliary signal to regularize a largely monolithic policy.

ForceVLA. ForceVLA serves as a stronger multimodal baseline with modality-specialized experts. It uses separate experts to process visual and force inputs and fuses them for action prediction. We keep the observation interface, action representation, and fine-tuning protocol consistent with the rest of the baselines. This comparison helps determine whether gains come simply from larger multimodal capacity or from explicitly modeling future contact dynamics.

D Task Descriptions and Success Criteria

USB Insertion. The robot grasps a USB connector, aligns it with the target port, and inserts it fully into the receptacle. This task requires accurate pose alignment, contact detection at the port entrance, and small corrective motions during insertion. A rollout is counted as successful if the USB plug enters the port with the correct orientation and reaches the fully seated configuration without being manually corrected or forcefully wedged at an incorrect angle. Failed partial insertions, obvious misalignment, or unstable contact that cannot progress to full insertion are counted as failures.

Gear Assembly. The robot picks and places a gear onto a target shaft or matching fixture and completes the mating process under contact constraints. The main challenge is resolving small pose errors during the final alignment stage, where the gear teeth and the shaft geometry create narrow feasible contact transitions. A rollout is successful if the gear is assembled into the designated target position and remains stably seated after release. Cases with visible tilting, incomplete seating, or contact deadlock are counted as failures.

Box Flipping. The robot reorients a box-like object by establishing and maintaining controlled contact with the table or another support surface. The task requires coordinated force modulation to prevent slipping or losing the object while transitioning between support modes. A rollout is successful if the object reaches the specified target orientation and remains stable at the end of execution. Any trial with object drop, uncontrolled slip, or incorrect final orientation is labeled as failure.

Board Wiping. The robot moves a tool or contact surface across a board while maintaining stable contact and appropriate motion coverage. Unlike insertion tasks, this task emphasizes sustained surface interaction rather than a single terminal contact event. A rollout is successful if the robot preserves continuous task-relevant contact with the board and completes the prescribed wiping motion over the target region without losing contact or deviating substantially from the intended path. Rollouts with premature lift-off, unstable chatter, or incomplete coverage are counted as failures.

VGA Plugging. The robot aligns a VGA connector with the socket and performs a plug-in motion under tight geometric tolerance and strong contact sensitivity. Compared with USB insertion, this task typically involves a wider connector body and stricter alignment requirements, making failure recovery more difficult once the connector is skewed. A rollout is successful if the connector is inserted into the VGA port to the target depth and remains securely engaged at the end of the rollout. Misaligned insertion attempts, shallow engagement, or jamming outside the valid socket configuration are counted as failures.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: Please see the summarized contributions in Section 1.

Guidelines:

- The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Please see Section A.

Guidelines:

- The answer [\[N/A\]](#) means that the paper has no limitation while the answer [\[No\]](#) means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The paper should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[N/A\]](#)

Justification: The paper does not present theorems or formal proofs; it gives a model formulation and training objective only.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The method, training objectives, model components, experimental setup, baselines, and implementation details are described in Section 4.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The paper reports real-world experiments, but the code and collected demonstrations are not released at submission time.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Training details, hardware setup, data collection, evaluation tasks, baselines, and success criteria are described in Sections 4.1, 4, 4.1, and D.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: The paper reports success rates over multiple real-world rollouts, but does not include confidence intervals, error bars, or statistical significance tests.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).

- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: The training hardware, batch size, optimizer, learning rate schedule, number of iterations, and approximate training time are reported in Section 4.1.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The manuscript describes a robotics method evaluated with teleoperated robot demonstrations and does not involve prohibited data collection, deception, or undisclosed human-subject experimentation.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section A discusses potential benefits for robotic assembly and assistive manipulation, as well as safety risks from incorrect contact reasoning during deployment.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.

- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release a trained model, dataset, or other high-risk asset.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [N/A]

Justification: The manuscript cites prior models and methods academically but does not release or repackage third-party assets.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.

- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: The paper does not introduce or release new code, models, or datasets.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not include crowdsourcing or human-subject experiments.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The work does not involve human-subject research.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.

- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs are not used as part of the core methodology. The proposed method, experiments, and conclusions do not depend on LLM-generated components.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.